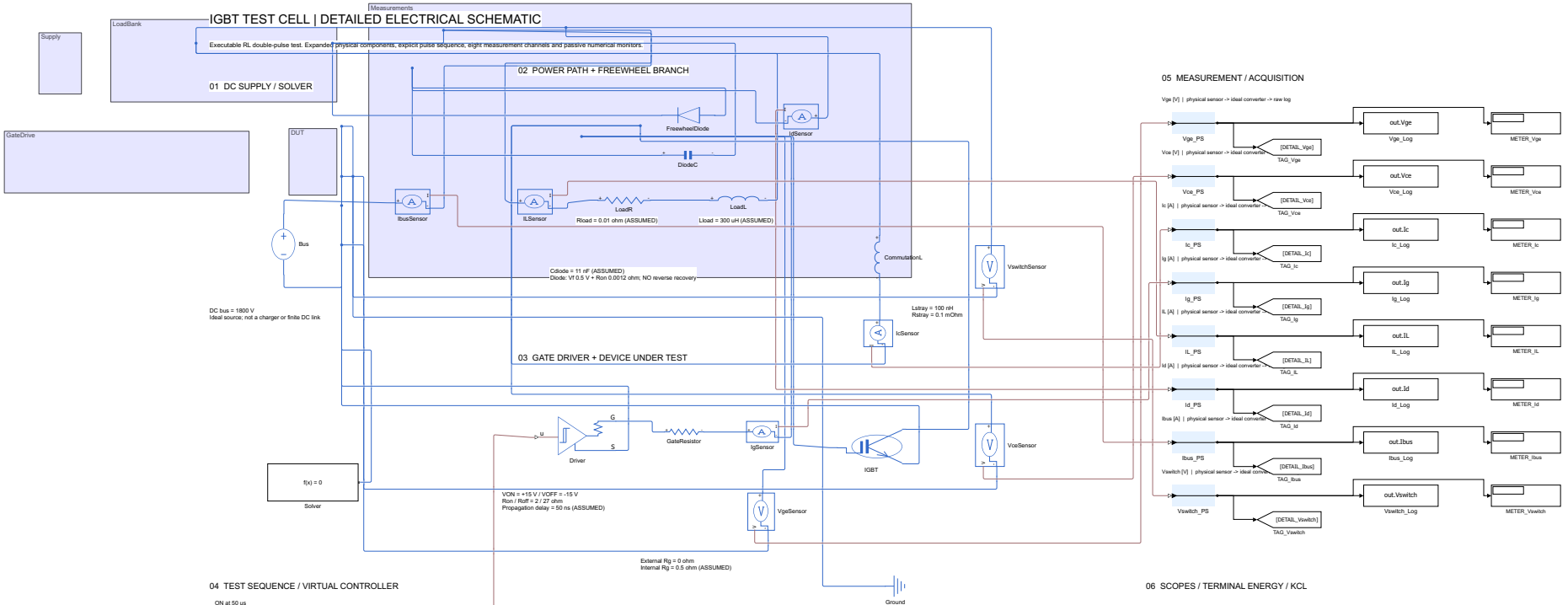

IGBT TEST CELL | DETAILED ELECTRICAL SCHEMATIC
Executable RL double-pulse test. Expanded physical components, explicit pulse sequence, eight measurement channels and passive numerical monitors.
Supply
LoadBank
01 DC SUPPLY / SOLVER
GateDrive
DUT
Measurements
02 POWER PATH + FREEWHEEL BRANCH
FreewheelDiode
DiodeC
IdSensor
IbusSensor
ILSensor
LoadR
LoadL
Rload = 0.01 ohm (ASSUMED)
Lload = 300 uH (ASSUMED)
Bus
CommutationL
Cdiode = 11 nF (ASSUMED)
Diode: Vf 0.5 V + Ron 0.0012 ohm; NO reverse recovery
VswitchSensor
Lstray = 100 nH
Rstray = 0.1 mOhm
IcSensor
DC bus = 1800 V
Ideal source; not a charger or finite DC link
03 GATE DRIVER + DEVICE UNDER TEST
G
S
u
Driver
GateResistor
IgSensor
IGBT
VceSensor
f(x) = 0
Solver
VON = +15 V / VOFF = -15 V
Ron / Roff = 2 / 27 ohm
Propagation delay = 50 ns (ASSUMED)
VgeSensor
External Rg = 0 ohm
Internal Rg = 0.5 ohm (ASSUMED)
Ground
05 MEASUREMENT / ACQUISITION
Vge [V] | physical sensor -> ideal converter -> raw log
Vge_PS
[DETAIL_Vge]
TAG_Vge
out.Vge
Vge_Log
METER_Vge
Vce [V] | physical sensor -> ideal converter
Vce_PS
[DETAIL_Vce]
TAG_Vce
out.Vce
Vce_Log
METER_Vce
Ic [A] | physical sensor -> ideal converter
Ic_PS
[DETAIL_Ic]
TAG_Ic
out.Ic
Ic_Log
METER_Ic
Ig [A] | physical sensor -> ideal converter
Ig_PS
[DETAIL_Ig]
TAG_Ig
out.Ig
Ig_Log
METER_Ig
IL [A] | physical sensor -> ideal converter
IL_PS
[DETAIL_IL]
TAG_IL
out.IL
IL_Log
METER_IL
Id [A] | physical sensor -> ideal converter
Id_PS
[DETAIL_Id]
TAG_Id
out.Id
Id_Log
METER_Id
Ibus [A] | physical sensor -> ideal converter
Ibus_PS
[DETAIL_Ibus]
TAG_Ibus
out.Ibus
Ibus_Log
METER_Ibus
Vswitch [V] | physical sensor -> ideal converter
Vswitch_PS
[DETAIL_Vswitch]
TAG_Vswitch
out.Vswitch
Vswitch_Log
METER_Vswitch
04 TEST SEQUENCE / VIRTUAL CONTROLLER
ON at 50 us
OFF at 300 us
Pulse_edge_1
ON at 400 us
Pulse_edge_2
OFF at 500 us
Pulse_edge_3
Pulse_edge_4
Command
CommandPS
[DETAIL_Vge]
Gate_waveforms_input_1
[DETAIL_Ig]
Gate_waveforms_input_2
Gate_waveforms
[DETAIL_Vce]
Switching_waveforms_input_1
[DETAIL_Ic]
Switching_waveforms_input_2
[DETAIL_IL]
Switching_waveforms
[DETAIL_Id]
Freewheel_waveforms_input_1
[DETAIL_Vswitch]
Freewheel_waveforms_input_2
[DETAIL_Ibus]
Freewheel_waveforms_input_3
Freewheel_waveforms
ON-1: 50..300 us | OFF: 300..400 us | ON-2: 400..500 us\nActual command edges above; driver includes propagation delay.\nOpen scope
06 SCOPES / TERMINAL ENERGY / KCL
[DETAIL_Vce]
Power_Vce
[DETAIL_Ic]
Power_Ic
x
Collector_terminal_power
1
s
Collector_terminal_energy
out.CollectorTerminalEnergy
Terminal_energy_log
Integral of Vce*Ic is collector TERMINAL energy/init includes conduction and reactive transfer; not calibrated pure Eon/Eoff.
[DETAIL_IL]
KCL_IL
[DETAIL_Ic]
KCL_Ic
[DETAIL_Id]
KCL_Id
KCL_residual
out.KCLResidual
KCL_log
KCL = IL - Ic - Id. Ideal sensor model; no ADC quantisation or noise.\nAuxiliary AC/DC, 24 V supply, optical isolation, CAN/USB and interlocks\nare not represented as hardware in this fixture. Hardware decision: INCONCLUSIVE.
Representative CM1500HC-66XB; NOT identified Wabtec module
Equation-based full I-V and charge model; fixed 25 degC
Vth input 6.3 V; calibration Vce 2.6 V at 1500 A / 25 degC
Cies / Cres / Coss = 164 / 1.4 / 11 nF
Transit time 0.2 us (ASSUMED); dynamic calibration remains incomplete